WORKED SOLUTION

STEP I 2001, Question 8

Pure • Second-Order Linear Equations

STEP I 2001, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q8

The key idea

Each known solution, substituted into the equation, gives one equation linking the two unknown functions $p(x)$ and $q(x)$ at each value of x. Two solutions give two simultaneous linear equations, which we solve for p and q exactly as we would for two numbers. In the final part, the linearity proved in the middle part lets us replace the two given solutions with simpler combinations of them, and the trigonometric identities make those combinations very simple indeed.

Solution

Finding p and q

For $y = x$ we have $y' = 1$ and $y'' = 0$, so substituting into $(*)$ gives

$$p(x) + x\,q(x) = 0. \tag{1}$$

For $y = 1 - x^2$ we have $y' = -2x$ and $y'' = -2$, so

$$-2 - 2x\,p(x) + (1 - x^2)\,q(x) = 0. \tag{2}$$

From (1), $p(x) = -x\,q(x)$. Substituting this into (2),

$$-2 + 2x^2 q(x) + (1 - x^2)q(x) = 0 \quad \Longrightarrow \quad (1 + x^2)\,q(x) = 2.$$

Since $1 + x^2 > 0$ we may divide, so $q(x) = 2(1 + x^2)^{-1}$, and then (1) gives $p(x) = -x\,q(x) = -2x(1 + x^2)^{-1}$, as required. (We never divided by x, so this holds for every real x, including $x = 0$.)

Any combination $ax + b(1 - x^2)$ is a solution

Write $L[y] = y'' + p\,y' + q\,y$. Differentiation is linear, so if $y = ay_1 + by_2$ with a, b constant, then

$$\begin{aligned} L[ay_1 + by_2] &= a(y_1'' + p\,y_1' + q\,y_1) + b(y_2'' + p\,y_2' + q\,y_2) \\ &= a\,L[y_1] + b\,L[y_2]. \end{aligned}$$

Taking $y_1 = x$ and $y_2 = 1 - x^2$, both $L[y_1]$ and $L[y_2]$ are zero (that is what it means for them to satisfy $(*)$), so $L[ax + b(1 - x^2)] = 0$ for all constants a and b. If you prefer to see it directly: $y = ax + b(1 - x^2)$ has $y' = a - 2bx$ and $y'' = -2b$, and

$$\begin{aligned} y'' + p\,y' + q\,y &= -2b - \frac{2x(a - 2bx)}{1 + x^2} + \frac{2(ax + b - bx^2)}{1 + x^2} \\ &= \frac{-2b(1 + x^2) - 2ax + 4bx^2 + 2ax + 2b - 2bx^2}{1 + x^2} = 0. \end{aligned}$$

The final part

The argument we just gave used nothing about the particular p and q: for *any* equation of the form $(*)$, a combination $ay_1 + by_2$ of two solutions is again a solution. So if $\cos^2(\frac{1}{2}x^2)$ and $\sin^2(\frac{1}{2}x^2)$ both satisfy $(*)$, so do their sum and difference,

$$\cos^2(\tfrac{1}{2}x^2) + \sin^2(\tfrac{1}{2}x^2) = 1, \qquad \cos^2(\tfrac{1}{2}x^2) - \sin^2(\tfrac{1}{2}x^2) = \cos(x^2),$$

by $\cos^2\theta + \sin^2\theta = 1$ and $\cos^2\theta - \sin^2\theta = \cos 2\theta$ with $\theta = \frac{1}{2}x^2$.

Ask yourself Why is it fair to work with 1 and $\cos(x^2)$ instead of the functions we were given?

Because the step is reversible: the given functions are $\frac{1}{2}(1 + \cos x^2)$ and $\frac{1}{2}(1 - \cos x^2)$, so p and q make both of them solutions exactly when they make both 1 and $\cos(x^2)$ solutions. We lose no information, and the derivatives become much easier.

The solution $y = 1$. Here $y' = y'' = 0$, so $(*)$ reduces to $q(x) \cdot 1 = 0$, giving

$$q(x) = 0.$$

The solution $y = \cos(x^2)$. By the chain rule $y' = -2x\sin(x^2)$, and by the product rule

$$y'' = -2\sin(x^2) - 4x^2\cos(x^2).$$

With $q = 0$, $(*)$ becomes

$$-2\sin(x^2) - 4x^2\cos(x^2) - 2x\sin(x^2)\,p(x) = 0.$$

Wherever $x\sin(x^2) \neq 0$ we can divide, and

$$p(x) = -\frac{\sin(x^2) + 2x^2\cos(x^2)}{x\sin(x^2)} = -\frac{1}{x} - 2x\cot(x^2).$$

So $p(x) = -\dfrac{1}{x} - 2x\cot(x^2)$ and $q(x) = 0$.

At the points where $x\sin(x^2) = 0$ (that is, $x = 0$ and $x = \pm\sqrt{n\pi}$) the coefficient of p vanishes, so the equation says nothing about p there; indeed our p blows up at these points. This is not a slip: at such points both given functions have zero derivative, so they cannot be two "independent" solutions of a well-behaved equation of the form $(*)$, and p must be singular there.

Checks

- As an independent check of the first part, the *Wronskian* $W = y_1y_2' - y_2y_1'$ of two solutions satisfies $W' = -pW$. With $y_1 = x$, $y_2 = 1 - x^2$ we get $W = -2x^2 - (1 - x^2) = -(1 + x^2)$, so $-p = W'/W = 2x/(1 + x^2)$, agreeing with our p. For the final part, the solutions 1 and $\cos(x^2)$ give $W = -2x\sin(x^2)$, and $W'/W = 1/x + 2x\cot(x^2) = -p$ again.
- Numerically, at $x = 0.9$ the two given functions, substituted into $(*)$ with derivatives computed by finite differences, force $p(0.9) \approx -2.82465$ and $q(0.9) \approx 0$, matching $-1/0.9 - 1.8\cot(0.81) \approx -2.82465$.

Common slips

- Differentiating $\cos^2(\frac{1}{2}x^2)$ directly, twice, and getting lost in the algebra. Double-angle identities first, then differentiate.
- Dividing (1) by x to get $q = -p/x$; this works, but then you must deal with $x = 0$ separately. Eliminating p instead avoids the issue.
- Forgetting to say where the formula for p in the final part is valid.

Beyond STEP

This question is a glimpse of the theory of second-order linear differential equations: solutions form a two-dimensional vector space, any two independent solutions determine the equation, and the Wronskian (with Abel's formula $W' = -pW$) measures independence. Points where the Wronskian vanishes, like $x = \pm\sqrt{n\pi}$ here, are exactly where the coefficients must become singular. Boyce and DiPrima, *Elementary Differential Equations*, develops all of this carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.